Collective Intelligence 2nd Assignment

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LevelBasedForaging

December 21, 2025

Research Questions

- Does discrete communication improve coordination and learning performance in `LevelBasedForaging-v0`?
- How does vocabulary size affect outcomes?
  - $V = 4$ (learned discrete symbols)
  - $V = 1$ (constant message, sanity check)
- How do critic choices affect learning?
  - MAPPO critic (centralized)
  - IPPO critic (decentralized)

Methods: Policies and Critic

Policies

- No communication:
  - Per-agent observation with shared trunk and linear action head
  - Categorical action distribution over 6 discrete actions
- Discrete communication:
  - Agents sample messages each step
  - Message is one-hot encoded and concatenated to next-step observation
  - Messages are swapped between agents and reset to 0 at episode termination

Critic

- MultiAgentMLP producing per-agent value estimates
- MAPPO: centralized critic
- IPPO: decentralized critic

Training and Metrics

Training Pipeline

- Data collection: `SyncDataCollector` and batched rollouts
- Storage: `ReplayBuffer`, sampling without replacement
- Objective: PPO with GAE
- Optimization: Adam, gradient clipping
- Training loop: flatten over team dimension; expand `done`/`terminated` shapes for PPO loss

Primary performance metric

- Mean episode return per training iteration
- Also plotted as rolling mean (window size $= 10$)

Experiments

1. **MAPPO**
   - No communication
   - Communication ($V = 4$)
   - Communication ($V = 1$)
2. **IPPO**
   - No communication
   - Communication ($V = 4$)
   - Communication ($V = 1$)

Results: Learning Curves

- MAPPO: no-comm vs $V = 4$ vs $V = 1$
- IPPO: no-comm vs $V = 4$ vs $V = 1$

To report

- Is $V = 4$ mean episode reward imrpoving?
- Does $V = 1$ match the no communication configuration?
- Are there any differences between MAPPO and IPPO?

MAPPO Learning Curve Plot

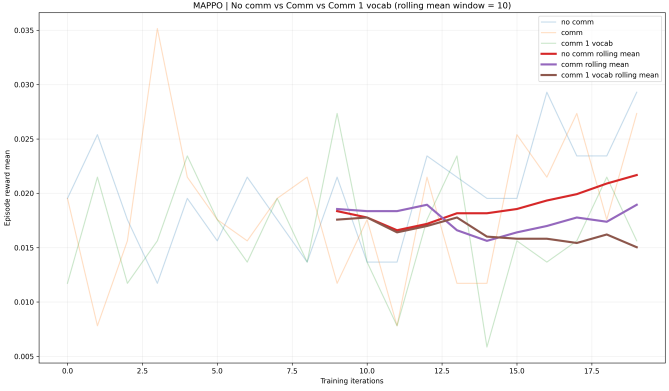


Figure: Learning curves of the MAPPO experiments

IPPO Learning Curve Plot

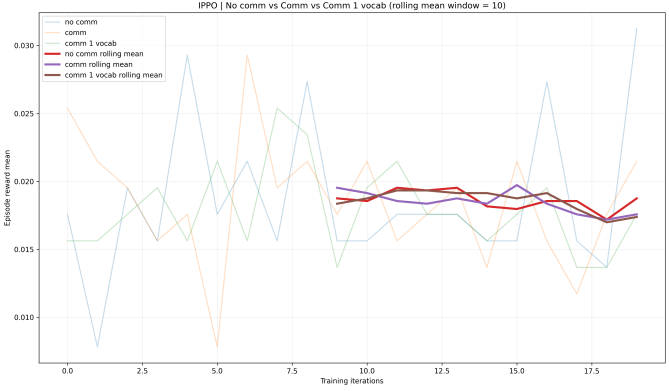


Figure: Learning curves of the IPPO experiments

Results: Entropy and Token Probability

- MAPPO: $V = 4$
- IPPO: $V = 4$

To report

- Is entropy decreasing?
- Do agents have a token preference?
- Are there any differences between MAPPO and IPPO?

MAPPO Entropy Plot

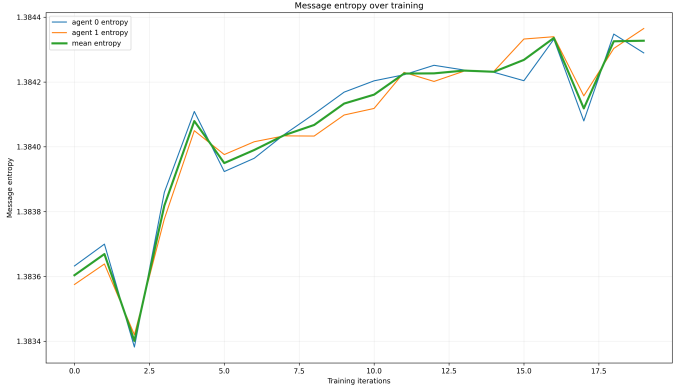


Figure: Message entropies of the MAPPO experiments

IPPO Entropy Plot

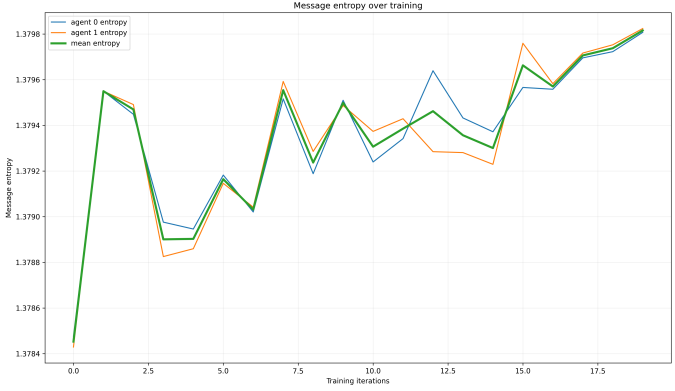


Figure: Message entropies of the IPPO experiments

MAPPO Token Probability Plot

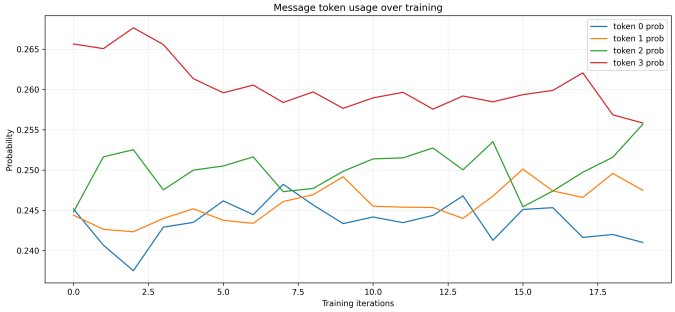


Figure: Message probabilities of the MAPPO experiments

IPPO Token Probabiltiy Plot

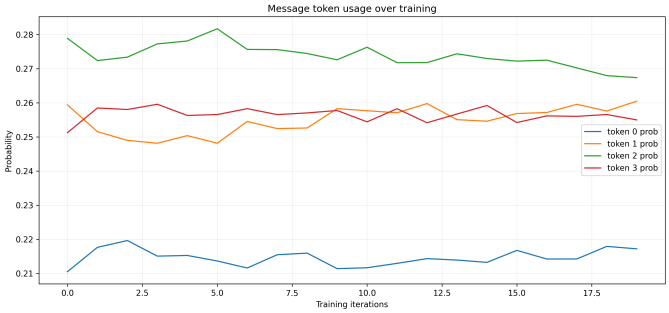


Figure: Message probabilities of the IPPO experiments

Failure Cases & Limitations

Failure Cases

- No performance gain from communication
- Messages ignored by policy
- Only single-seed runs
- Instability from reward sparsity or coordination difficulty

Implementation limitations

- Only 2 agents, message passing is hard-coded swap
- Only discrete messages, no continuous protocol
- No LSTM policy

Conclusions

Main takeaway

- As it is apparent from the plots, training essentially does nothing, there must be some fault in the code that escaped my attention despite my best efforts to uncover it

Next steps

- Deeply analyze why no learning occurs
- Run multiple random seeds, report mean $\pm$ std with confidence intervals
- Implementing LSTM policy
- Include more evaluation metrics